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TAPE ATTACHING APPARATUS

Abstract

Examples of the present disclosure relate to a tape attaching apparatus of which an installation space is reduced, with productivity of tape attaching work improved, and accuracy of a tape attachment position is improved. Examples of the present disclosure include a tape attaching apparatus including a tape supply unit that supplies a tape, an electrode plate supply unit that supplies an electrode plate, and a rotating head that receives the tape supplied from the tape supply unit and rotates to attach the tape to the electrode plate supplied from the electrode plate supply unit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims the benefit of priority from Korean Patent Application No. 10-2024-0027217, filed on Feb. 26, 2024 in the Korean Intellectual Property Office, the entire disclosure of which being incorporated herein by reference.

BACKGROUND

1. Field of the Invention

[0002] Examples of the present disclosure relates to a tape attaching apparatus.

2. Discussion of Related Art

[0003] Generally, the demand for portable electronic products such as, e.g., laptop computers, video cameras, portable phones, and the like, is increasing, and robots, electric vehicles, and the like are commercialized in earnest. Thus, research on high-performance secondary batteries capable of repeated charging and discharging is actively conducted.

[0004] A secondary battery includes a case and an electrode assembly accommodated inside the case. The electrode assembly includes a pair of electrode plates, and a separator interposed between the pair of electrode plates. Typically, each, or at least one, of the electrode plates includes a foil-shaped electrode current collector and an electrode active material with which a surface of the electrode current collector is coated.

[0005] An uncoated portion not coated with the electrode active material may be formed on the electrode plate. A tape may be attached to the uncoated portion to reduce or prevent short circuit.

SUMMARY OF THE INVENTION

[0006] An aspect of embodiments of the present disclosure is directed to providing a tape attaching apparatus of which an installation space is reduced or minimized, that has an improved productivity of tape attaching work, and an improved accuracy of a tape attachment position.

[0007] However, some technical objectives to be achieved by examples of the present disclosure are not limited to the above-described objectives, and other objectives, which are not described above, will be clearly understood by those skilled in the art through the following description of the disclosure.

[0008] According to an example aspect of the present disclosure, a tape attaching apparatus includes a tape supply unit configured to supply a tape, an electrode plate supply unit configured to supply an electrode plate, and a rotating head that receives the tape supplied from the tape supply unit and rotates to attach the tape to the electrode plate supplied from the electrode plate supply unit.

[0009] The rotating head may receive the tape from the tape supply unit at a tape transfer point, and rotate in one direction to attach the tape to the electrode plate at a tape attachment point.

[0010] After the tape is attached to the electrode plate at the tape attachment point, the rotating head may further rotate in the one direction, which is the same direction as above, to return to the tape transfer point.

[0011] The tape may include an adhesive surface coated with an adhesive, and a non-adhesive surface which is opposite to the adhesive surface that is not coated with the adhesive, and the rotating head may be configured to adsorb the non-adhesive surface at the tape transfer point and stop the adsorption of the non-adhesive surface at the tape attachment point.

[0012] The rotating head may push the tape toward the electrode plate so that the adhesive surface

adheres to the electrode plate at the tape attachment point.

[0013] The rotating head may be provided as, or include, a plurality of rotating heads, and the plurality of rotating heads may be arranged at, e.g., substantially equiangular intervals with respect to the same rotation center.

[0014] The tape attaching apparatus may further include a rotation frame that supports the plurality of rotating heads and rotates about the rotation center.

[0015] The tape transfer point may be provided as, or include, a plurality of tape transfer points, the tape attachment point may be provided as, or include, a plurality of tape attachment points, and the number of tape transfer points and the number of tape attachment points may be the same, and the number of rotating heads may be a multiple of the number of tape transfer points and the number of tape attachment points, e.g., the number of rotating heads may be twice the number of tape transfer points and the number of tape attachment points.

[0016] The tape may be transferred to one of the plurality of rotating heads from the tape supply unit at one of the plurality of tape transfer points, transferred to the electrode plate from the one rotating head at one of the plurality of tape attachment points, and attached to the electrode plate, and the one tape transfer point and the one tape attachment point may be spaced apart from each other by about 180° with respect to a rotation center of the rotating head.

[0017] A size of the tape transferred to one of the plurality of rotating heads at one of the plurality of tape transfer points may be different from a size of the tape transferred to another of the plurality of rotating heads at another of the plurality of tape transfer points.

[0018] The tape supply unit may be provided as, or include, a plurality of tape supply units in, e.g., a one-to-one correspondence with the tape transfer points, and the electrode plate supply unit may be provided as, or include, one electrode plate supply unit.

[0019] The tape supply unit may include a tape supply roller that is configured to substantially seamlessly and continuously supply the tape, and a tape cutter that is configured to cut the tape supplied from the tape supply roller to a desired or preset length.

[0020] The tape supply unit may further include a tape gripper that is configured to grip an end portion of the tape fed from the tape supply roller.

[0021] In a state in which the tape gripper grips the tape, the rotating head may adsorb the tape, and in a state in which the tape is adsorbed to the rotating head, the tape cutter may cut the tape.

[0022] The tape supply unit may further include a tape tension roller that is configured to pull the tape so that tension of the tape is substantially maintained.

[0023] The electrode plate supply unit may include an electrode plate supply roller that is configured to substantially seamlessly and continuously supply the electrode plate, and an electrode plate support that supports the electrode plate at the tape attachment point.

[0024] The electrode plate supply unit may further include an electrode plate stopping roller that is configured to stop movement of the electrode plate passing through the tape attachment point when the tape is attached to the electrode plate.

[0025] The rotating head may rotate after adsorbing the tape at the tape transfer point, may move in a radial direction from a rotation center of the rotating head at the tape attachment point to attach the tape to the electrode plate, may substantially stop the adsorbing of the tape, and may return toward the rotation center.

[0026] The electrode plate supply unit may further include an electrode plate tension roller that pulls the electrode plate so that tension of the electrode plate is substantially maintained.

[0027] The tape attachment point may be provided as, or include, a plurality of tape attachment points, the electrode plate supply roller may be provided as, or include, one electrode plate supply roller, the electrode plate support may be provided as, or include, a plurality of electrode plate supports provided in a one-to-one correspondence with the tape attachment points, and the plurality of electrode plate supports may be spaced apart from each other on a movement path of the electrode plate supplied from the electrode plate supply roller.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] The drawings submitted with this specification illustrate example embodiments of the present disclosure, and further describe aspects and features of the present disclosure together with the detailed description thereof. Thus, the present disclosure should not be construed as being limited to the drawings:

[0029] FIG. 1 is a diagram of a tape attaching apparatus, according to an example embodiment of the present disclosure;

[0030] FIG. 2 is an enlarged view of a tape supply unit of FIG. 1, which is a view illustrating a state in which a rotating head rotates to a tape transfer point, according to an example embodiment of the present disclosure;

[0031] FIG. 3 is a view illustrating a state in which an end of a tape is gripped by a tape gripper of FIG. 2, according to an example embodiment of the present disclosure;

[0032] FIG. 4 is a view illustrating a state in which the rotating head of FIG. 3 moves in a radial direction and the end of the tape is adsorbed to the rotating head, according to an example embodiment of the present disclosure;

[0033] FIG. 5 is a view illustrating a state in which a tape cutter of FIG. 4 cuts the end of the tape, according to an example embodiment of the present disclosure;

[0034] FIG. 6 is a view illustrating a state in which the rotating head adsorbs a tape piece and moves toward the center of rotation, according to an example embodiment of the present disclosure;

[0035] FIG. 7 is an enlarged view of part A of FIG. 1, which is a cross-sectional view illustrating a state in which the rotating head approaches an electrode plate so that the tape piece adsorbed to the rotating head is attached to the electrode plate, according to an example embodiment of the present disclosure;

[0036] FIG. 8 is a plan view illustrating an example of the electrode plate to which the tape piece is attached by the tape attaching apparatus, according to an example embodiment of the present disclosure;

[0037] FIG. 9 is a diagram of a tape attaching apparatus, according to an example embodiment of the present disclosure;

[0038] FIG. 10 is a plan view illustrating an example of the electrode plate to which the tape piece is attached by the tape attaching apparatus, according to an embodiment of the present disclosure;

[0039] FIG. 11 is a diagram of a tape attaching apparatus according to an example embodiment of the present disclosure; and

[0040] FIG. 12 is a plan view illustrating an example of the electrode plate to which the tape piece is attached by the tape attaching apparatus according to an example embodiment of the present disclosure.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0041] Herein, some example embodiments of the present disclosure will be described, in further detail, with reference to the accompanying drawings. The terms or words used in this specification and claims should not be construed as being limited to the usual or dictionary meaning and should be interpreted as meaning and concept consistent with the technical idea of the present disclosure based on the principle that the inventor can be his/her own lexicographer to appropriately define the concept of the term.

[0042] The example embodiments described in this specification and the configurations shown in the drawings are provided as some example embodiments of the present disclosure and do not represent all of the technical ideas, aspects, and features of the present disclosure. Accordingly, it is to be understood that there may be various equivalents and modifications that may replace or

modify the example embodiments described herein at the time of filing this application.

[0043] It is to be understood that when an element or layer is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it may be directly on, connected, or coupled to the other element or layer or one or more intervening elements or layers may also be present. When an element or layer is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. For example, when a first element is described as being “coupled” or “connected” to a second element, the first element may be directly coupled or connected to the second element or the first element may be indirectly coupled or connected to the second element via one or more intervening elements.

[0044] In the figures, dimensions of the various elements, layers, etc. may be exaggerated for clarity of illustration. The same reference numerals designate the same or like elements. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Further, the use of “may” when describing embodiments of the present disclosure relates to “one or more embodiments of the present disclosure.” Expressions, such as “at least one of” and “any one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. When phrases such as “at least one of A, B, and C,” “at least one of A, B, or C,” “at least one selected from a group of A, B, and C,” or “at least one selected from among A, B, and C” are used to designate a list of elements A, B, and C, the phrase may refer to any and all suitable combinations or a subset of A, B, and C, such as A, B, C, A and B, A and C, B and C, or A and B and C. As used herein, the terms “use,” “using,” and “used” may be considered synonymous with the terms “utilize,” “utilizing,” and “utilized,” respectively. As used herein, the terms “substantially,” “about,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent variations in measured or calculated values that would be recognized by those of ordinary skill in the art.

[0045] When the terms “about” or “substantially” are used in this specification in connection with a numerical value, it is intended that the associated numerical value include a tolerance of $\pm 10\%$ around the stated numerical value. The expression “up to” includes amounts of zero to the expressed upper limit and all values therebetween. When ranges are specified, the range includes all values therebetween such as increments of 0.1%.

[0046] It is to be understood that, although the terms “first,” “second,” “third,” etc. may be used herein to describe various elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are used to distinguish one element, component, region, layer, or section from another element, component, region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of example embodiments.

[0047] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It is to be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” or “over” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (e.g., rotated about 90 degrees or at other orientations), and the spatially relative descriptors used herein should be interpreted accordingly.

[0048] The terminology used herein is for the purpose of describing example embodiments of the present disclosure and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a” and “an” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It is to be further understood that the terms “includes,” “including,”

“comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0049] Also, any numerical range disclosed and/or recited herein is intended to include all sub-ranges of the same numerical precision subsumed within the recited range. For example, a range of “1.0 to 10.0” is intended to include all subranges between (and including) the recited minimum value of 1.0 and the recited maximum value of 10.0, that is, having a minimum value equal to or greater than 1.0 and a maximum value equal to or less than 10.0, such as, for example, 2.4 to 7.6. Any maximum numerical limitation recited herein is intended to include all lower numerical limitations subsumed therein, and any minimum numerical limitation recited in this specification is intended to include all higher numerical limitations subsumed therein. Accordingly, Applicant reserves the right to amend this specification, including the claims, to expressly recite any sub-range subsumed within the ranges expressly recited herein.

[0050] References to two compared elements, features, etc. as being “the same” may mean that they are “substantially the same.” Thus, the phrase “substantially the same” may include a case having a deviation that is considered low in the art, for example, a deviation of 5% or less. In addition, when a certain parameter is referred to as being uniform in a given region, it may mean that it is uniform in terms of an average.

[0051] Throughout the specification, unless otherwise stated, each element may be singular or plural.

[0052] When an arbitrary element is referred to as being arranged (or located or positioned) on the “above (or below)” or “on (or under)” a component, it may mean that the arbitrary element is placed in contact with the upper (or lower) surface of the component and may also mean that another component may be interposed between the component and any arbitrary element arranged (or located or positioned) on (or under) the component.

[0053] In addition, it is to be understood that when an element is referred to as being “coupled,” “linked,” or “connected” to another element, the elements may be directly “coupled,” “linked,” or “connected” to each other, or one or more intervening elements may be present therebetween, through which the element may be “coupled,” “linked,” or “connected” to another element. In addition, when a part is referred to as being “electrically coupled” to another part, the part may be directly electrically connected to another part or one or more intervening parts may be present therebetween such that the part and the “another” part are indirectly electrically connected to each other.

[0054] Throughout the specification, when “A and/or B” is stated, it means A, B, or A and B, unless otherwise stated. That is, “and/or” includes any or all combinations of a plurality of items enumerated. When “C to D” is stated, it means C or more and D or less, unless otherwise specified.

[0055] The terms used in the present specification are for describing embodiments of the present disclosure and are not intended to limit the present disclosure.

[0056] FIG. 1 is a diagram of a tape attaching apparatus according to an example embodiment of the present disclosure.

[0057] FIG. 2 is an enlarged view of a tape supply unit of FIG. 1, which is a view illustrating a state in which a rotating head rotates to a tape transfer point.

[0058] FIG. 3 is a view illustrating a state in which an end of a tape is gripped by a tape gripper of FIG. 2.

[0059] FIG. 4 is a view illustrating a state in which the rotating head of FIG. 3 moves in a radial direction and the end of the tape is adsorbed to the rotating head.

[0060] FIG. 5 is a view illustrating a state in which a tape cutter of FIG. 4 cuts the end of the tape.

[0061] FIG. 6 is a view illustrating a state in which the rotating head adsorbs a tape piece and moves toward the center of rotation.

[0062] FIG. 7 is an enlarged view of part A of FIG. 1, which is a cross-sectional view illustrating a state in which the rotating head approaches an electrode plate so that the tape piece adsorbed to the rotating head is attached to the electrode plate.

[0063] FIG. 8 is a plan view illustrating an example of the electrode plate to which the tape piece is attached by the tape attaching apparatus according to an example embodiment of the present disclosure.

[0064] Referring to FIGS. 1 to 8, a tape attaching apparatus 100 according to an example embodiment of the present disclosure is or includes an apparatus for attaching a tape 20 to an electrode plate 11A. In detail, the tape attaching apparatus 100 may automatically attach, to an uncoated portion 16 of an electrode plate 11A, a tape piece 25 of the tape 20 formed by cutting a portion of the tape 20 that continues substantially seamlessly.

[0065] The electrode plate 11A, along with a separator, constitutes an electrode assembly. The electrode plate 11A includes an electrode current collector 12 in the form of a foil and an electrode active material layer 14 applied to a surface of the electrode current collector 12 and containing an electrode active material. Since the electrode plate 11A is not coated with the electrode active material layer 14, the uncoated portion 16 through which the electrode current collector 12 is exposed may be formed. An electrode tab (not illustrated) may be welded to the uncoated portion 16. The electrode tab may be electrically connected to a terminal of a secondary battery.

[0066] When the electrode plate 11A is a negative electrode plate, the electrode current collector 12 may be a metal foil made of or including, for example, at least one of copper, a copper alloy, nickel, and a nickel alloy. The electrode active material layer 14 may contain, for example, graphite or carbon.

[0067] When the electrode plate 11A is a positive electrode plate, the electrode current collector 12 may be a metal foil made of or including, for example, at least one of aluminum and an aluminum alloy. The electrode active material layer 14 may contain, for example, a transition metal oxide.

[0068] The tape attaching apparatus 100 includes a tape supply unit 120, an electrode plate supply unit 140, and a rotating head 110. The tape supply unit 120 supplies the tape 20, specifically, the cut tape piece 25 of tape 20. The electrode plate supply unit 140 supplies the electrode plate 11A.

[0069] The rotating head 110 receives the tape 20, that is, the tape piece 25 of the tape 20, supplied from the tape supply unit 120, and rotates to attach the tape piece 25 of the tape 20 to the electrode plate 11A supplied from the electrode plate supply unit 140.

[0070] The rotating head 110 may be provided as, or include, a plurality of rotating heads 110. The plurality of rotating heads 110 may be arranged at, e.g., equiangular intervals with respect to the same rotation center RX. For example, the rotating heads 110, e.g., four rotating heads 110, may be arranged at, e.g., about 90° intervals with respect to a circle having as a center the rotation center RX. In other examples, the rotating heads 110 may be arranged at other angular intervals with respect to a circle having as a center the rotation center RX.

[0071] The tape attaching apparatus 100 may further include a rotation frame 101 that is configured to support the plurality of rotating heads 110 and to rotate about the rotation center RX. The rotation frame 101 may rotate in one direction. For example, the rotation frame 101 and the plurality of rotating heads 110 may rotate about the rotation center RX in a counterclockwise direction.

[0072] Each, or at least one, of the rotating heads 110 may move away from the rotation center RX and in a backward direction. In other words, the rotating head 110 may move in a radial direction and toward the rotation center RX.

[0073] A negative pressure may be formed at a distal end 112 of one or more of the rotating heads 110. Thus, the tape piece 25 of the tape 20 may be adsorbed to the distal end 112 of the rotating head 110. Detailed descriptions of an apparatus configured to rotate and drive the rotation frame 101, an apparatus configured to move the rotating head 110 in the radial direction and toward the rotation center RX, and an apparatus configured to form the negative pressure at the distal end 112

of the rotating head **110** will be omitted.

[0074] As illustrated in FIG. **1**, the rotating head **110** may receive the tape piece **25** of the tape **20** from the supply unit **120** at a tape transfer point TF and may rotate in one direction to be attached to the electrode plate **11A** at a tape attachment point TA. The tape transfer point TF and the tape attachment point TA may be spaced apart from each other at, e.g., about 180° intervals with respect to the rotation center RX. In other examples, the tape transfer point TF and the tape attachment point TA may be spaced apart from each other at other angular intervals with respect to the rotation center RX.

[0075] The rotation frame **101** may rotate at a constant speed. However, at a moment that one rotating head **110** among the plurality of rotating heads **110** is positioned at the tape transfer point TF and another rotating head **110** opposite to the one rotating head **110** is positioned at the tape attachment point TA, the rotation of the rotation frame **101** and the plurality of rotating heads **110** supported thereon may temporarily stop.

[0076] At a moment that the rotation is stopped, the tape piece **25** of the tape **20** may be adsorbed to the distal end **112** of the one rotating head **110** at the tape transfer point TF, and the tape piece **25** of the tape **20** may be attached to the uncoated portion **16** of the electrode plate **11A** at the tape attachment point TA.

[0077] For example, in a state in which the one rotating head **110** among the plurality of rotating heads **110** rotationally moves to the tape transfer point TF, the tape piece **25** of the tape **20** is attached to the distal end **112** of the rotating head **110**, and the rotating head **110** rotates by about 180° in a counterclockwise direction and moves to the tape attachment point TA. Further, the tape piece **25** of the tape **20** attached to the rotating head **110** is transferred and attached to the uncoated portion **16** of the electrode plate **11A** at the tape attachment point TA, and the rotating head **110** rotates by about 180° in the counterclockwise direction as well and moves to the tape transfer point TF.

[0078] The tape supply unit **120** may include a tape supply roller **121**, a tape gripper **123**, tape stopping rollers **126** and **127**, a tape cutter **128**, and a tape tension roller **130**. The tape supply roller **121** unwinds the tape **20** wound in the form of a roll and substantially seamlessly and continuously supplies the tape **20**.

[0079] The tape **20** may be made of or include a synthetic resin that is an insulating material. The tape **20** includes both side surfaces opposite to each other. One of the side surfaces of the tape **20** may be or include an adhesive surface coated with an adhesive, and the other side surface opposite to the adhesive surface may be or include a non-adhesive surface that is not coated with an adhesive.

[0080] The tape gripper **123** may be configured to grip an end portion **22** of the tape **20** fed from the tape supply roller **121**. The tape gripper **123** may include a plurality of fingers configured to move in a folding direction in which the tape gripper **123** is folded to grip the end portion **22** of the tape **20**, or to move in an unfolding direction opposite to the folding direction.

[0081] The tape cutter **128** is configured to cut the end portion **22** of the tape **20** supplied from the tape supply roller **121** to a preset length. The end portion **22** of the tape **20** cut by the tape cutter **128** may become the tape piece **25** of the tape **20** that is adsorbed to the distal end **112** of the rotating head **110**.

[0082] The tape stopping rollers **126** and **127** are configured to stop movement of the tape **20** when the end portion **22** of the tape **20** is cut by the tape cutter **128**. The tape stopping rollers **126** and **127** may be provided as, or include, a pair of tape stopping rollers **126** and **127**, and the pair of tape stopping rollers **126** and **127** may be substantially close to each other.

[0083] While the end portion **22** of the tape **20** passes between the pair of tape stopping rollers **126** and **127** which are substantially close to each other, when the pair of tape stopping rollers **126** and **127** are stopped, the end portion **22** of the tape **20** is stopped without movement. In this case, the tape cutter **128** may cut the end portion **22** of the tape **20**.

[0084] The tape tension roller **130** pulls the tape **20** so that the tension of the tape **20** is maintained. The tape tension roller **130** may be on a path of the tape **20** from the tape supply roller **121** to a space between the pair of tape stopping rollers **126** and **127**.

[0085] The tape supply unit **120** may further include an upstream tension guide roller **133** on an upstream side of the tape tension roller **130** and a downstream tension guide roller **134** on a downstream side of the tape tension roller **130** along the movement path of the tape **20**. The tape **20** supplied from the tape supply roller **121** may sequentially pass through the upstream tension guide roller **133**, the tape tension roller **130**, and the downstream tension guide roller **134** and may enter between the pair of tape stopping rollers **126** and **127**.

[0086] When the tension of the tape **20** decreases, the tape tension roller **130** may move in a direction that increases the path of the tape **20** and thus increase the tension of the tape **20**, and when the tension of the tape **20** increases, the tape tension roller **130** may move in a direction that decreases the path of the tape **20** and thus decrease the tension of the tape **20**. Accordingly, the tension of the tape **20** can be maintained to be substantially uniform.

[0087] For example, as illustrated in FIG. **3**, when the tension of the tape **20** decreases, the tape tension roller **130** may move away from the upstream and downstream tension guide rollers **133** and **134**, and when the tension of the tape **20** increases, the tape tension roller **130** may move in a direction closer to the upstream and downstream tension guide rollers **133** and **134**. Since the tape tension roller **130** moves and the substantially constant tension of the tape **20** can be maintained, while the pair of tape stopping rollers **126** and **127** are stopped to cut the end portion **22** of the tape **20**, the tape supply roller **121** may substantially continuously rotate without stopping.

[0088] Hereinafter, a process of transferring the tape piece **25** obtained by cutting the tape **20** to the rotating head **110** will be sequentially described. First, as illustrated in FIG. **2**, when the rotating head **110** rotates and then stops at the tape transfer point TF, the end portion **22** of the tape **20** may be interposed between the pair of tape stopping rollers **126** and **127**, and a distal end of the tape **20** may protrude from the tape gripper **123**.

[0089] Next, as illustrated in FIG. **3**, the tape gripper **123** may grip the distal end of the tape **20**. When the pair of tape stopping rollers **126** and **127** rotate so that the distal end of the tape **20** approaches the tape gripper **123** and then stop, the tape gripper **123** may grip the distal end of the tape **20**.

[0090] Alternatively, the tape gripper **123** may move closer to the pair of tape stopping rollers **126** and **127** to grip the distal end of the tape **20**. Then, the tape gripper **123** may move away from the pair of tape stopping rollers **126** and **127** in turn, and the pair of tape stopping rollers **126** and **127** may rotate so that the tape **20** moves at the same speed as a moving speed of the tape gripper **123**.

[0091] Next, as illustrated in FIG. **4**, the rotating head **110** moves away from the rotation center RX so that the distal end **112** of the rotating head **110** comes into contact with the end portion **22** of the tape **20**. Further, a negative pressure is formed at the distal end **112** of the rotating head **110** to adsorb the end portion **22** of the tape **20**. The rotating head **110** may adsorb the end portion **22** of the tape **20** in a state in which the tape gripper **123** grips the end portion **22** of the tape **20**.

[0092] Next, as illustrated in FIG. **5**, in a state in which the end portion **22** of the tape **20** is gripped by the tape gripper **123** and adsorbed to the rotating head **110**, the tape cutter **128** is operated to cut the end portion **22** of the tape **20** into the tape piece **25**. In this case, a non-adhesive surface **28** (see FIG. **6**) of the tape piece **25** may be adsorbed to the distal end **112** of the rotating head **110** to face the distal end **112** of the rotating head **110**.

[0093] Next, as illustrated in FIG. **6**, the plurality of fingers of the tape gripper **123** are unfolded to separate the tape piece **25** from the tape gripper **123**, and the rotating head **110** moves to an original position thereof, that is, toward to the rotation center RX, while adsorbing the tape piece **25**. The tape cutter **128** also returns to an original position thereof.

[0094] Referring back to FIG. **1**, the electrode plate supply unit **140** may include an electrode plate supply roller **141**, an electrode plate collection roller **143**, an electrode plate support **145**, upstream

electrode plate stopping rollers **148** and **149**, downstream electrode plate stopping rollers **153** and **154**, a first electrode plate tension roller **156**, and a second electrode plate tension roller **162**. The electrode plate supply roller **141** unwinds the electrode plate **11A** wound in the form of a roll and substantially seamlessly and continuously supplies the unwound electrode plate **11A**.

[0095] With respect to FIGS. **7** and **8**, the uncoated portion **16** not coated with the electrode active material layer **14** and a mixture portion **19** coated with the electrode active material layer **14** may be formed on the electrode plate **11A** alternately in a longitudinal direction of the electrode plate **11A**. The electrode plate collection roller **143** collects the electrode plate **11A** having the tape piece **25** attached to the uncoated portion **16** by winding the electrode plate **11A** it into a roll shape.

[0096] The electrode plate support **145** supports the electrode plate **11A** at the tape attachment point TA (see FIG. **1**). The rotating head **110**, which adsorbs and rotates the tape piece **25** to the distal end **112** and moves to the tape attachment point TA, pushes the tape piece **25** toward the electrode plate **11A** so that an adhesive surface **27** of the tape piece **25** adheres to the uncoated portion **16** of the electrode plate **11A**. The rotating head **110** moves in the radial direction and away from the rotation center RX illustrated in FIG. **1**, the adhesive surface **27** of the tape piece **25** is pressed against and comes into close contact with the uncoated portion **16** of the electrode plate **11A** supported on the electrode plate support **145**, and thus the tape piece **25** may be substantially firmly attached to the uncoated portion **16**.

[0097] The electrode plate stopping rollers **148**, **149**, **153**, and **154** stop the movement of the electrode plate **11A** passing through the tape attachment point TA when the tape piece **25** is attached to the electrode plate **11A**. The electrode plate stopping rollers may include the upstream electrode plate stopping rollers **148** and **149** arranged on an upstream side of the electrode plate support **145** along a movement path of the electrode plate **11A**, and the downstream electrode plate stopping rollers **153** and **154** arranged on a downstream side of the electrode plate support **145** along the movement path of the electrode plate **11A**.

[0098] The upstream electrode plate stopping rollers **148** and **149** may be provided as, or include, a pair of upstream electrode plate stopping rollers **148** and **149**, and the downstream electrode plate stopping rollers **153** and **154** may be provided as, or include, a pair of downstream electrode plate stopping rollers **153** and **154**. The pair of upstream electrode plate stopping rollers **148** and **149** may be substantially close to each other, and the pair of downstream electrode plate stopping rollers **153** and **154** may be substantially close to each other.

[0099] When the electrode plate **11A** passes between the pair of upstream electrode plate stopping rollers **148** and **149** substantially close to each other and the pair of downstream electrode plate stopping rollers **153** and **154** substantially close to each other, and when the pair of upstream electrode plate stopping rollers **148** and **149** and the pair of downstream electrode plate stopping rollers **153** and **154** are stopped, a portion of the electrode plate **11A** between the upstream electrode plate stopping rollers **148** and **149** and the downstream electrode plate stopping rollers **153** and **154** stops as well. In this case, the rotating head **110** may attach the tape piece **25** to the uncoated portion **16** of the electrode plate **11A**.

[0100] The electrode plate tension rollers **156** and **162** illustrated in FIG. **1** pull the electrode plate **11A** so that tension of the electrode plate **11A** is maintained. The electrode plate tension roller may include a first electrode plate tension roller **156** located on an upstream side of the electrode plate support **145** and the upstream electrode plate stopping rollers **148** and **149** along the movement path of the electrode plate **11A** and a second electrode plate tension roller **162** located on a downstream side of the electrode plate support **145** and the downstream electrode plate stopping rollers **153** and **154** along the movement path of the electrode plate **11A**.

[0101] The electrode plate supply unit **140** may further include an upstream first electrode plate tension guide roller **158** on an upstream side of the first electrode plate tension roller **156** and a downstream first electrode plate tension guide roller **159** on a downstream side of the first electrode plate tension roller **156** along the movement path of the electrode plate **11A**.

[0102] The electrode plate **11A** supplied from the electrode plate supply roller **141** may sequentially pass through the upstream first electrode plate tension guide roller **158**, the first electrode plate tension roller **156**, the downstream first electrode plate tension guide roller **159**, and a gap between the pair of upstream electrode plate stopping rollers **148** and **149**, and then move onto the electrode plate support **145**.

[0103] In this section, when a tensile force of the electrode plate **11A** decreases, the first electrode plate tension roller **156** may move in a direction that increases the path of the electrode plate **11A**, to increase the tensile force of the electrode plate **11A**, and when the tensile force of the electrode plate **11A** increases, the first electrode plate tension roller **156** may move in a direction that decreases the path of the electrode plate **11A**, to decrease the tensile force of the electrode plate **11A**. Accordingly, the tensile force of the electrode plate **11A** supplied from the electrode plate supply roller **141** and moving to the electrode plate support **145** may be maintained to be substantially uniform.

[0104] For example, when the tensile force of the electrode plate **11A** decreases, the first electrode plate tension roller **156** may move away from the upstream and downstream first electrode plate tension guide rollers **158** and **159**, and when the tensile force of the electrode plate **11A** increases, the first electrode plate tension roller **156** may move in a direction closer to the upstream and downstream first electrode plate tension guide rollers **158** and **159**.

[0105] Since the first electrode plate tension roller **156** moves and the constant tensile force of the electrode plate **11A** can be maintained, the electrode plate supply roller **141** may substantially continuously rotate without stopping to attach the tape piece **25** to the electrode plate **11A** while the pair of upstream electrode plate stopping rollers **148** and **149** and the pair of downstream electrode plate stopping rollers **153** and **154** are stopped.

[0106] The electrode plate supply unit **140** may further include an upstream second electrode plate tension guide roller **163** on an upstream side of the second electrode plate tension roller **162** and a downstream second electrode plate tension guide roller **164** on a downstream side of the second electrode plate tension roller **162**.

[0107] The electrode plate **11A**, on which the tape piece **25** is attached to the uncoated portion **16**, may pass between the pair of downstream electrode plate stopping rollers **153** and **154**, may sequentially pass through the upstream second electrode plate tension guide roller **163**, the second electrode plate tension roller **162**, and the downstream second electrode plate tension guide roller **164**, and may be collected by the electrode plate collection roller **143**.

[0108] In this section, when the tensile force of the electrode plate **11A** decreases, the second electrode plate tension roller **162** may move in a direction that increases the path of the electrode plate **11A**, to increase the tensile force of the electrode plate **11A**, and when the tensile force of the electrode plate **11A** increases, the second electrode plate tension roller **162** may move in a direction that decreases the path of the electrode plate **11A**, to decrease the tensile force of the electrode plate **11A**. Accordingly, the tension force of the electrode plate **11A** moving from the electrode plate support **145** toward the electrode plate collection roller **143** may be maintained to be substantially uniform.

[0109] For example, when the tensile force of the electrode plate **11A** decreases, the second electrode plate tension roller **162** may move away from the upstream and downstream second electrode plate tension guide rollers **163** and **164**, and when the tensile force of the electrode plate **11A** increases, the second electrode plate tension roller **162** may move in a direction closer to the upstream and downstream second electrode plate tension guide rollers **163** and **164**.

[0110] Since the second electrode plate tension roller **162** moves and the constant tensile force of the electrode plate **11A** can be maintained, the electrode plate collection roller **143** may continue to rotate without stopping to attach the tape piece **25** to the electrode plate **11A** while the pair of upstream electrode plate stopping rollers **148** and **149** and the pair of downstream electrode plate stopping rollers **153** and **154** are stopped.

[0111] Although not illustrated, an electrode tab welding unit, which welds an electrode tab (not illustrated) to the uncoated portion **16** of the electrode plate **11A**, may be on the movement path of the electrode plate **11A** between the electrode plate supply roller **141** and the first electrode plate tension roller **156**. In this case, at the tape attachment point TA, the tape piece **25** may be attached to the uncoated portion **16** to cover the uncoated portion **16** and a portion of the electrode tab welded to the uncoated portion **16**.

[0112] In a state in which the rotating head **110** that adsorbs the tape piece **25** moves to the tape attachment point TA, and the rotating head **110** and the uncoated portion **16** of the electrode plate **11A** are aligned in a radial direction of the rotation center RX, the rotating head **110** may move in the radial direction with respect to the rotation center RX to attach the tape piece **25** to the uncoated portion **16** of the electrode plate **11A**. The tape piece **25** may be attached to the electrode plate **11A** to cover the uncoated portion **16**.

[0113] Next, the rotating head **110** may stop adsorbing the tape piece **25** and return toward the rotation center RX. After the tape piece **25** is attached to the electrode plate **11A**, the rotating head **110** may further rotate in the same rotation direction as a case in which the rotating head **110** rotates to the tape attachment point TA to move toward the tape transfer point TF.

[0114] Although not illustrated in FIGS. **1** and **8**, the electrode tab welding unit, which welds the electrode tab (not illustrated) to the uncoated portion **16** of the electrode plate **11A**, may be on the movement path of the electrode plate **11A** from the electrode plate supply roller **141** toward the first electrode plate tension roller **156**. In this case, at the tape attachment point TA, the tape piece **25** may be attached to the uncoated portion **16** to cover the uncoated portion **16** along with the portion of the electrode tab welded to the uncoated portion **16**.

[0115] FIG. **9** is a diagram of a tape attaching apparatus according to another embodiment of the present disclosure, and FIG. **10** is a plan view illustrating an example of the electrode plate to which the tape piece is attached by the tape attaching apparatus according to the example embodiment of the present disclosure. Referring to FIGS. **9** and **10**, a tape attaching apparatus **200** according to the example embodiment of the present disclosure includes a plurality of tape supply units **220A** and **220B**, one electrode supply unit **240**, a plurality of rotating heads **210A** and **210B**, and a rotation frame **201**.

[0116] The tape supply units **220A** and **220B** supply the tape **20**, specifically, tape pieces **25A** and **25B** of the cut tape **20** illustrated in FIG. **10**. The tape supply unit includes a first tape supply unit **220A** and a second tape supply unit **220B**.

[0117] The first tape supply unit **220A** is configured to cut the substantially continuous tape **20** (see FIG. **2**) and to supply the first tape piece **25A** having the same size as the tape piece **25** illustrated in FIG. **8**. The second tape supply unit **220B** is configured to cut the tape **20** and to supply the second tape piece **25B** that is larger than the first tape piece **25A**. A width of the tape **20** cut into the second tape piece **25B** may be, e.g., larger than a width of the tape **20** cut into the first tape piece **25A**.

[0118] The rotating heads **210A** and **210B** are configured to receive the tape pieces **25A** and **25B** supplied from the tape supply units **220A** and **220B**, and rotate to attach the tape piece **25A** and **25B** to an electrode plate **11B** supplied from the electrode plate supply unit **240**.

[0119] The rotating head **210A** and **210B** may include the first rotating head **210A** that is configured to receive and adsorb the first tape piece **25A**, and the second rotating head **210B** that is configured to receive and adsorb the second tape piece **25B**. The first rotating head **210A** may be provided as, or include, a pair of first rotating heads **210A**, the second rotating head **210B** may be provided as, or include, a pair of second rotating heads **210B**, and thus a total of four rotating heads may be provided.

[0120] The pair of first rotating heads **210A** and the pair of second rotating heads **210B** may be arranged at equiangular intervals, that is, at about 90° intervals, with respect to the same rotation center RX, and may be arranged alternately. The pair of first rotating heads **210A** may be arranged

at about 180° intervals with respect to the rotation center RX, and the pair of second rotating heads **210B** may be arranged at about 180° intervals with respect to the rotation center RX.

[0121] The pair of first rotating heads **210A** may be arranged at about 180° intervals with respect to the rotation center RX, and the rotation frame **201** may support the plurality of rotating heads **210A** and **210B** and rotate about the rotation center RX. The rotation frame **201** may rotate in one direction. For example, the rotation frame **201** and the plurality of rotating heads **210A** and **210B** may rotate about the rotation center RX in a counterclockwise direction.

[0122] Each, or at least one, of the rotating heads **210A** and **210B** may move away from the rotation center RX and in the backward direction. In other words, the rotating heads **210A** and **210B** may move in the radial direction and toward the rotation center RX.

[0123] A negative pressure may be formed at distal ends **212** of the rotating heads **210A** and **210B**. Thus, the first tape piece **25A** may be adsorbed on the distal end **212** of the first rotating head **210A**, and the second tape piece **25B** may be adsorbed to the distal end **212** of the second rotating head **210B**.

[0124] The first rotating head **210A** may receive the first tape piece **25A** from the first tape supply unit **220A** at a first tape transfer point TF1 and rotate in one direction to attach the first tape piece **25A** to the first uncoated portion **16** of the electrode plate **11B** at a first tape attachment point TA1. The first tape transfer point TF1 and the first tape attachment point TA1 may be spaced apart from each other at about 180° intervals with respect to the rotation center RX.

[0125] The second rotating head **210B** may receive the second tape piece **25B** from the second tape supply unit **220A** at a second tape transfer point TF2 and rotate in one direction to attach the second tape piece **25B** to a second uncoated portion **17** of the electrode plate **11B** at a second tape attachment point TA2. The second tape transfer point TF2 and the second tape attachment point TA2 may be spaced apart from each other at about 180° intervals with respect to the rotation center RX.

[0126] The rotation frame **201** may rotate, e.g., at a constant or substantially constant speed. However, at a moment that the plurality of rotating heads **210A** and **210B** are positioned at the tape transfer points TF1 and TF2 or the tape attachment points TA1 and TA2, the rotation of the rotation frame **201** and the plurality of rotating heads **210A** and **210B** supported thereon may temporarily stop.

[0127] For example, at a moment that the rotation of the rotation frame **201** is stopped, the first tape piece **25A** is adsorbed to the distal end **212** of one of the pair of first rotating heads **210A** at the first tape transfer point TF1, and the first tape piece **25A** may be attached to the first uncoated portion **16** of the electrode plate **11B** from the other first rotating head **210A** at the first tape attachment point TA1.

[0128] Further, the second tape piece **25B** is adsorbed to the distal end **212** of one of the pair of second rotating heads **210B** at the second tape transfer point TF2, and the second tape piece **25B** may be attached to the second uncoated portion **17** of the electrode plate **11B** from the other second rotating head **210B** at the second tape attachment point TA2.

[0129] In a state in which one of the pair of first rotating heads **210A** rotates and moves to the first tape transfer point TF1, the first tape piece **25A** is attached to the distal end **212** of the first rotating head **210A**, and the first rotating head **210A** rotates by about 180° in a counterclockwise direction and moves to the first tape attachment point TA1. Further, the first tape piece **25A** attached to the first rotating head **210A** at the first tape attachment point TA1 is moved and attached to the first uncoated portion **16** of the electrode plate **11B**, and the first rotating head **210A** rotates by about 180° in a counterclockwise direction again and moves to the first tape transfer point TF1.

[0130] In a state in which one of the pair of second rotating heads **210B** rotates and moves to the second tape transfer point TF2, the second tape piece **25B** is attached to the distal end **212** of the second rotating head **210B**, and the second rotating head **210B** rotates by about 180° in a counterclockwise direction and moves to the second tape attachment point TA2. Further, the second

tape piece **25B** attached to the second rotating head **210B** at the second tape attachment point **TA2** is moved and attached to the second uncoated portion **17** of the electrode plate **11B**, and the second rotating head **210B** rotates by about 180° in a counterclockwise direction again and moves to the second tape transfer point **TF2**.

[0131] Since the first tape supply unit **220A** has the same configuration as the tape supply unit **120** included in the tape attaching apparatus **100** according to the example embodiment of the present disclosure, a duplicated description thereof will be omitted. The second tape supply unit **220B** has the same configuration as the tape supply unit **120** included in the tape attaching apparatus **100** according to the example embodiment of the present disclosure with a difference that the second tape supply unit **220B** supplies the second tape piece **25B** that is larger than the first tape piece **25A**, and thus a duplicated description thereof is largely omitted herein.

[0132] The electrode plate supply unit **240** may include an electrode plate supply roller **241**, an electrode plate collection roller **243**, electrode plate supports **245** and **255**, first electrode plate stopping rollers **248**, **249**, **253**, and **254**, second electrode plate stopping rollers **258**, **259**, **263**, and **264**, a first electrode plate tension roller **266**, a second electrode plate tension roller **272**, and a third electrode plate tension roller **276**. The electrode plate supply roller **241** unwinds the electrode plate **11B** wound in the form of a roll and substantially seamlessly and continuously supplies the unwound electrode plate **11B**.

[0133] The electrode plate **11B** may be provided with the first uncoated portion **16** and the second uncoated portion **17** not coated with the electrode active material layer **14** and the mixture portion **19** coated with the electrode active material layer **14**. A planar size of the second uncoated portion **17** may be larger than a planar size of the first uncoated portion **16**. The planar size of the first uncoated portion **16** may correspond to a planar size of the first tape piece **25A**, and the planar size of the second uncoated portion **17** may correspond to a planar size of the second tape piece **25B**.

[0134] The first uncoated portion **16** and the second uncoated portion **17** may be spaced apart from each other in a longitudinal direction of the electrode plate **11B** with the mixture portion **19** interposed therebetween. The first uncoated portion **16** and the second uncoated portion **17** may be arranged alternately.

[0135] The electrode plate collection roller **243** winds the electrode plate **11B**, in which the first tape piece **25A** is attached to the first uncoated portion **16** and the second tape piece **25B** is attached to the second uncoated portion **17**, into a roll shape and collects the wound electrode plate **11B**.

[0136] The electrode plate supports **245** and **255** may be provided as, or include, a pair of electrode plate supports **245** and **255**. The first electrode plate support **245** supports the electrode plate **11B** at the first tape attachment point **TA1**. The second electrode plate support **255** supports the electrode plate **11B** at the second tape attachment point **TA2**.

[0137] The first rotating head **210A**, which adsorbs the first tape piece **25A** at the distal end **212** and rotates to move to the first tape attachment point **TA1**, pushes the first tape piece **25A** toward the electrode plate **11B** so that an adhesive surface of the first tape piece **25A** adheres to the first uncoated portion **16** of the electrode plate **11B**. The first rotating head **210A** moves in the radial direction and away from the rotation center **RX**, the adhesive surface of the first tape piece **25A** is pressed against and in close contact with the first uncoated portion **16** of the electrode plate **11B** supported on the first electrode plate support **245**, and thus the first tape piece **25A** may be substantially firmly attached to the first uncoated portion **16**.

[0138] The second rotating head **210A**, which adsorbs the second tape piece **25B** at the distal end **212** and rotates to move to the second tape attachment point **TA2**, pushes the second tape piece **25A** toward the electrode plate **11B** so that an adhesive surface of the second tape piece **25B** adheres to the second uncoated portion **17** of the electrode plate **11B**. The second rotating head **210B** moves in the radial direction and away from the rotation center **RX**, the adhesive surface of the second tape piece **25B** is pressed against and in close contact with the second uncoated portion **17** of the electrode plate **11B** supported on the second electrode plate support **255**, and thus the second tape

piece **25B** may be substantially firmly attached to the second uncoated portion **17**.

[0139] The first electrode plate stopping rollers **248**, **249**, **253**, and **254** stop the movement of the electrode plate **11B** passing through the first tape attachment point **TA1** when the first tape piece **25A** is attached to the electrode plate **11B**. The first electrode plate stopping rollers may include the first upstream electrode plate stopping rollers **248** and **249** arranged on an upstream side of the first electrode plate support **245** along a movement path of the electrode plate **11B** and the first downstream electrode plate stopping rollers **253** and **254** arranged on a downstream side of the first electrode plate support **245** along the movement path of the electrode plate **11B**.

[0140] The first upstream electrode plate stopping rollers **248** and **249** may be provided as, or include, a pair of first upstream electrode plate stopping rollers **248** and **249**, and the first downstream electrode plate stopping rollers **253** and **254** may be provided as, or include, a pair of first downstream electrode plate stopping rollers **253** and **254**. The pair of first upstream electrode plate stopping rollers **248** and **249** may be substantially close to each other, and the pair of first downstream electrode plate stopping rollers **253** and **254** may be substantially close to each other.

[0141] When the electrode plate **11B** passes between the pair of first upstream electrode plate stopping rollers **248** and **249** substantially close to each other and the pair of first downstream electrode plate stopping rollers **253** and **254** being substantially close to each other, and when the pair of first upstream electrode plate stopping rollers **248** and **249** and the pair of first downstream electrode plate stopping rollers **253** and **254** are stopped, a portion of the electrode plate **11B** between the first upstream electrode plate stopping rollers **248** and **249** and the first downstream electrode plate stopping rollers **253** and **254** stops as well. In this case, the first rotating head **210A** may attach the first tape piece **25A** to the first uncoated portion **16** of the electrode plate **11B**.

[0142] The second electrode plate stopping rollers **258**, **259**, **263**, and **264** stop the movement of the electrode plate **11B** passing through the second tape attachment point **TA2** when the second tape piece **25B** is attached to the electrode plate **11B**. The second electrode plate stopping rollers may include the second upstream electrode plate stopping rollers **258** and **259** arranged on an upstream side of the second electrode plate support **255** along the movement path of the electrode plate **11B** and the second downstream electrode plate stopping rollers **263** and **264** arranged on a downstream side of the second electrode plate support **255** along the movement path of the electrode plate **11B**.

[0143] The second upstream electrode plate stopping rollers **258** and **259** may be provided as, or include, a pair of second upstream electrode plate stopping rollers **258** and **259**, and the second downstream electrode plate stopping rollers **263** and **264** may be provided as, or include, a pair of second downstream electrode plate stopping rollers **263** and **264**. The pair of second upstream electrode plate stopping rollers **258** and **259** may be substantially close to each other, and the pair of second downstream electrode plate stopping rollers **263** and **264** may be substantially close to each other.

[0144] When the electrode plate **11B** passes between the pair of second upstream electrode plate stopping rollers **258** and **259** which are substantially close to each other and the pair of second downstream electrode plate stopping rollers **263** and **264** which are substantially close to each other, and when the pair of second upstream electrode plate stopping rollers **258** and **259** and the pair of second downstream electrode plate stopping rollers **263** and **264** are stopped, a portion of the electrode plate **11B** between the second upstream electrode plate stopping rollers **258** and **259** and the second downstream electrode plate stopping rollers **263** and **264** stops as well. In this case, the second rotating head **210B** may attach the second tape piece **25B** to the second uncoated portion **17** of the electrode plate **11B**.

[0145] The electrode plate tension rollers **266**, **272** and **276** pull the electrode plate **11B** so that tension of the electrode plate **11B** is maintained. The electrode plate tension rollers may include the first electrode plate tension roller **266** located on an upstream side of the first electrode plate support **245** and the first upstream electrode plate stopping rollers **248** and **249** along the movement

path of the electrode plate **11B**, the second electrode plate tension roller **272** located between the first downstream electrode plate stopping rollers **253** and **254** and the second upstream electrode plate stopping rollers **258** and **259** along the movement path of the electrode plate **11B**, and the third electrode plate tension roller **276** located on a downstream side of the second electrode plate support **255** and the second downstream electrode plate stopping rollers **263** and **264** along the movement path of the electrode plate **11B**.

[0146] The electrode plate supply unit **240** may further include an upstream first electrode plate tension guide roller **268** on an upstream side of the first electrode plate tension roller **266** and a downstream first electrode plate tension guide roller **269** on a downstream side of the first electrode plate tension roller **266** along the movement path of the electrode plate **11B**.

[0147] The electrode plate **11B** supplied from the electrode plate supply roller **241** may sequentially pass through the upstream first electrode plate tension guide roller **268**, the first electrode plate tension roller **266**, the downstream first electrode plate tension guide roller **269**, and a gap between the pair of first upstream electrode plate stopping rollers **248** and **249**, and then move onto the first electrode plate support **245**.

[0148] In this section, when a tensile force of the electrode plate **11B** decreases, the first electrode plate tension roller **266** may move in a direction that increases the path of the electrode plate **11B**, to increase the tensile force of the electrode plate **11B**, and when the tensile force of the electrode plate **11B** increases, the first electrode plate tension roller **266** may move in a direction that decreases the path of the electrode plate **11B**, to decrease the tensile force of the electrode plate **11B**. Accordingly, the tensile force of the electrode plate **11B** supplied from the electrode plate supply roller **241** and moving to the first electrode plate support **245** may be maintained to be substantially uniform.

[0149] Since the first electrode plate tension roller **266** moves and the constant tensile force of the electrode plate **11B** can be maintained, the electrode plate supply roller **241** may substantially continuously rotate without stopping to attach the first tape piece **25A** to the electrode plate **11B** while the pair of first upstream electrode plate stopping rollers **248** and **249** and the pair of first downstream electrode plate stopping rollers **253** and **254** are stopped.

[0150] The electrode plate supply unit **240** may further include an upstream second electrode plate tension guide roller **273** on an upstream side of the second electrode plate tension roller **272** and a downstream second electrode plate tension guide roller **274** on a downstream side of the second electrode plate tension roller **272** along the movement path of the electrode plate **11B**.

[0151] In a section between the pair of first downstream electrode plate stopping rollers **253** and **254** and the pair of second upstream electrode plate stopping rollers **258** and **259**, when the tensile force of the electrode plate **11B** decreases, the second electrode plate tension roller **272** may move in a direction that increases the path of the electrode plate **11B**, to increase the tensile force of the electrode plate **11B**, and when the tensile force of the electrode plate **11B** increases, the second electrode plate tension roller **272** may move in a direction that decreases the path of the electrode plate **11B**, to decrease the tensile force of the electrode plate **11B**. Accordingly, the tensile force of the electrode plate **11B** from the first electrode plate support **245** to the second electrode plate support **255** can be maintained to be substantially uniform.

[0152] The electrode plate supply unit **240** may further include an upstream third electrode plate tension guide roller **278** on an upstream side of the third electrode plate tension roller **276** and a downstream third electrode plate tension guide roller **279** on a downstream side of the third electrode plate tension roller **276** along the movement path of the electrode plate **11B**.

[0153] The electrode plate **11B**, on which the second tape piece **25B** is attached to the second uncoated portion **17**, may pass between the pair of second downstream electrode plate stopping rollers **263** and **264**, may pass, e.g., sequentially pass, through the upstream third electrode plate tension guide roller **278**, the third electrode plate tension roller **276**, and the downstream third electrode plate tension guide roller **279**, and may be collected by the electrode plate collection

roller **243**.

[0154] In this section, when the tensile force of the electrode plate **11B** decreases, the third electrode plate tension roller **276** may move in a direction that increases the path of the electrode plate **11B** to increase the tensile force of the electrode plate **11B**, and when the tensile force of the electrode plate **11B** increases, the third electrode plate tension roller **276** may move in a direction that decreases the path of the electrode plate **11B** to decrease the tensile force of the electrode plate **11B**. Accordingly, the tension force of the electrode plate **11B** moving from the second electrode plate support **255** to the electrode plate collection roller **243** may be maintained to be substantially uniform.

[0155] Since the third electrode plate tension roller **276** moves and the constant tensile force of the electrode plate **11B** can be maintained, the electrode plate collection roller **243** may substantially continuously rotate without stopping to attach the second tape piece **25B** to the electrode plate **11B** while the pair of second upstream electrode plate stopping rollers **258** and **259** and the pair of second downstream electrode plate stopping rollers **263** and **264** are stopped.

[0156] The first rotating head **210A**, which adsorbs the first tape piece **25A**, may move to the first tape attachment point **TA1**, and the second rotating head **210B**, which adsorbs the second tape piece **25B**, may move to the second tape attachment point **TA2**, e.g., simultaneously or contemporaneously. In a state in which the first rotating head **210A** and the first uncoated portion **16** of the electrode plate **11B** are aligned in the radial direction of the rotation center **RX**, the first rotating head **210A** may move in the radial direction from the rotation center **RX** to attach the first tape piece **25A** to the first uncoated portion **16** of the electrode plate **11B**. The first tape piece **25A** may be attached to the electrode plate **11B** to cover the first uncoated portion **16**.

[0157] Further, in a state in which the second rotating head **210B** and the second uncoated portion **17** of the electrode plate **11B** are aligned in the radial direction of the rotation center **RX**, the second rotating head **210B** may move in the radial direction from the rotation center **RX** to attach the second tape piece **25B** to the second uncoated portion **17** of the electrode plate **11B**. The second tape piece **25B** may be attached to the electrode plate **11B** to cover the second uncoated portion **17**.

[0158] In an example, the rotating heads **210A** and **210B** may stop adsorbing the tape pieces **25A** and **25B** and return toward the rotation center **RX**. After the first tape piece **25A** is attached to the electrode plate **11B**, the first rotating head **210A** may further rotate in the same rotation direction as a case in which the first rotating head **210A** rotates to the first tape attachment point **TA1** to move toward the first tape transfer point **TF1**. After the second tape piece **25B** is attached to the electrode plate **11B**, the second rotating head **210B** may further rotate in the same rotation direction as a case in which the second rotating head **210B** rotates to the second tape attachment point **TA2** to move toward the second tape transfer point **TF2**.

[0159] Although not illustrated in FIGS. **9** and **10**, the electrode tab welding unit, which welds the electrode tab (not illustrated) to the uncoated portions **16** and **17** of the electrode plate **11B**, may be located on the movement path of the electrode plate **11B** from the electrode plate supply roller **241** toward the first electrode plate tension roller **266**. In this case, at the tape attachment points **TA1** and **TA2**, the tape pieces **25A** and **25B** may be attached to the uncoated portions **16** and **17** to cover the uncoated portion **16** and **17** along with the portion of the electrode tab welded to the uncoated portion **16** and **17**.

[0160] In the tape attaching apparatus **200** according to the other example embodiment of the present disclosure, the numbers of the plurality of tape transfer points **TF1** and **TF2** and the tape attachment points **TA1** and **TA2** are the same, and the number of rotating heads **210A** and **210B** is twice the number of tape transfer points **TF1** and **TF2** and the number of tape attachment points **TA1** and **TA2**.

[0161] The tape pieces **25A** and **25B** are transferred to one of the plurality of the rotating heads **210A** and **210B** from the tape supply units **220A** and **220B** at one of the plurality of tape transfer points **TF1** and **TF2** and are transferred and attached to the electrode plate **11B** from the one

rotating head at one of the plurality of tape attachment points TA1 and TA2. The one tape transfer point and the one tape attachment point are spaced apart from each other at about 180° intervals with respect to the plurality of rotation centers RX.

[0162] A size of the tape pieces 25A and 25B transferred to one of the plurality of rotating heads 210A and 210B at one of the plurality of tape transfer points TF1 and TF2 is different from a size of the tape pieces 25A and 25B transferred to the other of the plurality of rotating heads 210A and 210B at the other of the plurality of tape transfer points TF1 and TF2.

[0163] The plurality of tape supply units 220A and 220B are provided in a one-to-one correspondence with the tape transfer points TF1 and TF2, and the one electrode plate supply unit 240 is provided.

[0164] The plurality of tape attachment points TA1 and TA2 are provided, the one electrode plate supply roller 241 is provided, and the plurality of electrode plate supports 245 and 255 are provided in a one-to-one correspondence with the tape attachment points TA1 and TA2. The plurality of electrode plate supports 245 and 255 are spaced apart from each other on the movement path of the electrode plate 11B supplied from the electrode plate supply roller 241.

[0165] FIG. 11 is a diagram of a tape attaching apparatus according to a another embodiment of the present disclosure, and FIG. 12 is a plan view illustrating an example of the electrode plate to which the tape piece is attached by the tape attaching apparatus according to the example embodiment of the present disclosure. Referring to FIGS. 11 and 12, the tape attaching apparatus 300 according to the example embodiment of the present disclosure includes a plurality of tape supply units 320A, 320B, and 320C, and one electrode plate supply unit 340, a plurality of rotating heads 310A, 310B, and 310C, and a rotation frame 301.

[0166] The plurality of tape supply units 320A, 320B, and 320C supply the tape 20, specifically, tape pieces 25A, 25B, and 25C of the cut tape 20. The tape supply unit include the first tape supply unit 320A, the second tape supply unit 320B, and the third tape supply unit 320C.

[0167] The first tape supply unit 320A is configured to cut the substantially continuous tape 20 (see FIG. 2) and to supply the first tape piece 25A having the same size as the tape piece 25 illustrated in FIG. 8. The second tape supply unit 320B is configured to cut the tape 20 and to supply the second tape piece 25B that is larger than the first tape piece 25A. The third tape supply unit 320C is configured to cut the tape 20 and to supply the third tape piece 25C that is larger than the second tape piece 25B.

[0168] The width of the tape 20 cut into the second tape piece 25B may be greater than the width of the tape 20 cut into the first tape piece 25A, and a width of the tape 20 cut into the third tape piece 25C may be greater than the width of the tape 20 cut into the second tape piece 25B.

[0169] The rotating heads 310A, 310B, and 310C may be configured to receive the tape pieces 25A, 25B, and 25C supplied from the tape supply units 320A, 320B, and 320C and rotate to attach the tape piece 25A, 25B, and 25C to an electrode plate 11C supplied from the electrode plate supply unit 340.

[0170] The rotating heads 310A, 310B, and 310C may include the first rotating head 310A that is configured to receive and to adsorb the first tape piece 25A, a second rotating head 310B that is configured to receive and to adsorb the second tape piece 25B, and the third rotating head 310C that is configured to receive and to adsorb the third tape piece 25C. The first rotating head 310A may be provided as, or include, a pair of first rotating heads 310A, the second rotating head 310B may be provided as, or include, a pair of second rotating heads 310B, the third rotating head 310C may be provided as, or include, a pair of third rotating heads 310C, and a total of six rotating heads may be provided.

[0171] The pair of first rotating heads 310A, the pair of second rotating heads 310B, and the pair of third rotating heads 310C may be arranged at, e.g., equiangular intervals, in other words, about 60° intervals with respect to the same rotation center RX. The first rotating heads 310A, the second rotating heads 310B, and the third rotating heads 310C may be arranged sequentially. The pair of

first rotating heads **310A** may be arranged at about 180° intervals with respect to the rotation center RX, the pair of second rotating heads **310B** may be arranged at about 180° intervals with respect to the rotation center RX, and the pair of third rotating heads **310C** may be arranged at about 180° intervals with respect to the rotation center RX.

[0172] The rotation frame **301** may support the plurality of rotating heads **310A**, **310B**, and **310C** and rotate about the rotation center RX. The rotation frame **301** may rotate in one direction. For example, the rotation frame **301** and the plurality of rotating heads **310A**, **310B**, and **310C** may rotate about the rotation center RX in a counterclockwise direction.

[0173] Each, or at least one, of the rotating heads **310A**, **310B**, and **310C** may move away from the rotation center RX and in back towards the center RX. In other words, the rotating heads **310A**, **310B**, and **310C** may move in the radial direction and toward the rotation center RX.

[0174] A negative pressure may be formed at distal ends **312** of the rotating heads **310A**, **310B**, and **310C**. Thus, the first tape piece **25A** may be adsorbed to the distal end **312** of the first rotating head **310A**, the second tape piece **25B** may be adsorbed to the distal end **312** of the second rotating head **310B**, and the third tape piece **25C** may be adsorbed to the distal end **312** of the third rotating head **310C**.

[0175] The first rotating head **310A** may receive the first tape piece **25A** from the first tape supply unit **320A** at the first tape transfer point TF1 and rotate in one direction to attach the first tape piece **25A** to the first uncoated portion **16** of the electrode plate **11C** at the first tape attachment point TA1. The first tape transfer point TF1 and the first tape attachment point TA1 may be spaced apart from each other at about 180° intervals with respect to the rotation center RX.

[0176] The second rotating head **310B** may receive the second tape piece **25B** from the second tape supply unit **320B** at the second tape transfer point TF2 and rotate in one direction to attach the second tape piece **25B** to the second uncoated portion **17** of the electrode plate **11C** at the second tape attachment point TA2. The second tape transfer point TF2 and the second tape attachment point TA2 may be spaced apart from each other at about 180° intervals with respect to the rotation center RX.

[0177] The third rotating head **310C** may receive the third tape piece **25C** from the first tape supply unit **220C** at a third tape transfer point TF3 and rotate in one direction to attach the third tape piece **25C** to a third uncoated portion **18** of the electrode plate **11C** at a third tape attachment point TA3. The third tape transfer point TF3 and the third tape attachment point TA3 may be spaced apart from each other at about 180° intervals with respect to the rotation center RX.

[0178] The rotation frame **301** may rotate at a constant, or substantially constant, speed. However, at a moment that the plurality of rotating heads **310A**, **310B**, and **310C** are positioned at the tape transfer points TF1, TF2, and TF3 or the tape attachment points TA1, TA2, and TA3, the rotation of the rotation frame **301** and the plurality of rotating heads **310A**, **310B**, and **310C** supported thereon may temporarily stop.

[0179] For example, at a moment that the rotation of the rotation frame **301** is stopped, the first tape piece **25A** is adsorbed to the distal end **312** of one of the pair of first rotating heads **310A** at the first tape transfer point TF1, and the first tape piece **25A** may be attached to the first uncoated portion **16** of the electrode plate **11C** from the other first rotating head **310A** at the first tape attachment point TA1.

[0180] The second tape piece **25B** is adsorbed to the distal end **312** of one of the pair of second rotating heads **310B** at the second tape transfer point TF2, and the second tape piece **25B** may be attached to the second uncoated portion **17** of the electrode plate **11C** from the other second rotating head **310B** at the second tape attachment point TA2.

[0181] The third tape piece **25C** is adsorbed to the distal end **312** of one of the pair of third rotating heads **310C** at the third tape transfer point TF3, and the third tape piece **25C** may be attached to the third uncoated portion **18** of the electrode plate **11C** from the other third rotating head **310C** at the third tape attachment point TA3.

[0182] In a state in which one of the pair of first rotating heads **310A** rotates and moves to the first tape transfer point **TF1**, the first tape piece **25A** is attached to the distal end **312** of the first rotating head **310A**, and the first rotating head **310A** rotates by about 180° in a counterclockwise direction and moves to the first tape attachment point **TA1**. Further, the first tape piece **25A** attached to the first rotating head **310A** at the first tape attachment point **TA1** may be moved and attached to the first uncoated portion **16** of the electrode plate **11C**, and the first rotating head **310A** rotates by about 180° in a counterclockwise direction again and moves to the first tape transfer point **TF1**.

[0183] In a state in which one of the pair of second rotating heads **310B** rotates and moves to the second tape transfer point **TF2**, the second tape piece **25B** is attached to the distal end **312** of the second rotating head **310B**, and the second rotating head **310B** rotates by about 180° in a counterclockwise direction and moves to the second tape attachment point **TA2**. Further, the second tape piece **25B** attached to the second rotating head **310B** at the second tape attachment point **TA2** is moved and attached to the second uncoated portion **17** of the electrode plate **11C**, and the second rotating head **310B** rotates by about 180° in a counterclockwise direction again and moves to the second tape transfer point **TF2**.

[0184] In a state in which one of the pair of third rotating heads **310C** rotates and moves to the third tape transfer point **TF3**, the third tape piece **25C** is attached to the distal end **312** of the third rotating head **310C**, and the third rotating head **310C** rotates by about 180° in a counterclockwise direction and moves to the third tape attachment point **TA3**. Further, the third tape piece **25C** attached to the third rotating head **310C** at the third tape attachment point **TA3** is moved and attached to the third uncoated portion **18** of the electrode plate **11C**, and the third rotating head **310C** rotates by about 180° in a counterclockwise direction again and moves to the third tape transfer point **TF3**.

[0185] Since the first tape supply unit **320A** has the same configuration as the tape supply unit **120** included in the tape attaching apparatus **100** according to the example embodiment of the present disclosure, a duplicated description thereof will be omitted. The second tape supply unit **320B** and the third tape supply unit **320C** have the same configuration as the tape supply unit **120** included in the tape attaching apparatus **100** according to the example embodiment of the present disclosure illustrated in FIG. 1 with a difference that the second tape piece **25B** and the third tape piece **25C** have different sizes from the size of the first tape piece **25A**, and thus a duplicated description thereof is substantially omitted herein.

[0186] The electrode plate supply unit **340** may include an electrode plate supply roller **341**, an electrode plate collection roller **343**, electrode plate supports **345**, **355**, and **365**, first electrode plate stopping rollers **348**, **349**, **353**, and **354**, second electrode plate stopping rollers **358**, **359**, **363**, and **365**, third electrode plate stopping rollers **368**, **369**, **373**, and **374**, a first electrode plate tension roller **376**, a second electrode plate tension roller **382**, a third electrode plate tension roller **386**, and a fourth electrode plate tension roller **392**. The electrode plate supply roller **341** unwinds the electrode plate **11C** wound in the form of a roll and substantially seamlessly and continuously supplies the unwound electrode plate **11B**.

[0187] The electrode plate **11C** may be provided with, or include, the first uncoated portion **16**, the second uncoated portion **17**, the third uncoated portion **18** that is not coated with the electrode active material layer **14**, and the mixture portion **19** coated with the electrode active material layer **14**. The planar size of the second uncoated portion **17** may be greater than the planar size of the first uncoated portion **16**, and a planar size of the third uncoated portion **18** may be greater than the planar size of the second uncoated portion **17**. The planar size of the first uncoated portion **16** may correspond to the planar size of the first tape piece **25A**, the planar size of the second uncoated portion **17** may correspond to the planar size of the second tape piece **25B**, and the planar size of the third uncoated portion **18** may correspond to a planar size of the third tape piece **25C**.

[0188] The first uncoated portion **16**, the second uncoated portion **17**, and the third uncoated portion **18** may be spaced apart from each other in a longitudinal direction of the electrode plate

11C, with the mixture portion **19** interposed therebetween. The first uncoated portion **16**, the second uncoated portion **17**, and the third uncoated portion **18** may be arranged sequentially. [0189] The electrode plate collection roller **343** is configured to wind the electrode plate **11C** into a roll shape and to collect the electrode plate **11C** in which the first tape piece **25A** is attached to the first uncoated portion **16**, the second tape piece **25B** is attached to the second uncoated portion **17**, and the third tape piece **25C** is attached to the third uncoated portion **18**.

[0190] The electrode plate supports **345**, **355**, and **365** may be provided as, or include, three electrode plate supports **345**, **355**, and **365**. The first electrode plate support **345** supports the electrode plate **11C** at the first tape attachment point **TA1**. The second electrode plate support **355** supports the electrode plate **11C** at the second tape attachment point **TA2**. The third electrode plate support **365** supports the electrode plate **11C** at the third tape attachment point **TA3**.

[0191] The first rotating head **310A**, which adsorbs the first tape piece **25A** at the distal end **312** and rotates to move to the first tape attachment point **TA1**, pushes the first tape piece **25A** toward the electrode plate **11C** so that the adhesive surface of the first tape piece **25A** adheres to the first uncoated portion **16** of the electrode plate **11C**. The first rotating head **310A** moves in the radial direction and away from the rotation center **RX**, the adhesive surface of the first tape piece **25A** is pressed against and in close contact with the first uncoated portion **16** of the electrode plate **11C** supported on the first electrode plate support **345**, and thus the first tape piece **25A** may be substantially firmly attached to the first uncoated portion **16**.

[0192] The second rotating head **310B**, which adsorbs the second tape piece **25B** at the distal end **312** and rotates to move to the second tape attachment point **TA2**, pushes the second tape piece **25B** toward the electrode plate **11C** so that the adhesive surface of the second tape piece **25B** adheres to the second uncoated portion **17** of the electrode plate **11C**. The second rotating head **310B** moves in the radial direction and away from the rotation center **RX**, the adhesive surface of the second tape piece **25B** is pressed against and in close contact with the second uncoated portion **17** of the electrode plate **11C** supported on the second electrode plate support **355**, and thus the second tape piece **25B** may be substantially firmly attached to the second uncoated portion **17**.

[0193] The third rotating head **310C**, which is configured to adsorb the third tape piece **25C** at the distal end **312** and to rotate to move to the third tape attachment point **TA3**, pushes the third tape piece **25C** toward the electrode plate **11C** so that an adhesive surface of the third tape piece **25C** adheres to the third uncoated portion **18** of the electrode plate **11C**. The third rotating head **310C** moves in the radial direction and away from the rotation center **RX**, the adhesive surface of the third tape piece **25C** is pressed against and in close contact with the third uncoated portion **18** of the electrode plate **11C** supported on the third electrode plate support **365**, and thus the third tape piece **25C** may be substantially firmly attached to the third uncoated portion **18**.

[0194] The first electrode plate stopping rollers **348**, **349**, **353**, and **354** are configured to stop the movement of the electrode plate **11C** passing through the first tape attachment point **TA1** when the first tape piece **25A** is attached to the electrode plate **11C**. The first electrode plate stopping rollers may include the first upstream electrode plate stopping rollers **348** and **349** arranged on an upstream side of the first electrode plate support **345** along a movement path of the electrode plate **11C**, and the first downstream electrode plate stopping rollers **353** and **354** arranged on a downstream side of the first electrode plate support **345** along the movement path of the electrode plate **11C**.

[0195] The first upstream electrode plate stopping rollers **348** and **349** may be provided as, or include, a pair of first upstream electrode plate stopping rollers **248** and **249**, and the first downstream electrode plate stopping rollers **353** and **354** may be provided as, or include, a pair of first downstream electrode plate stopping rollers **253** and **254**. The pair of first upstream electrode plate stopping rollers **348** and **349** may be substantially close to each other, and the pair of first downstream electrode plate stopping rollers **353** and **354** may be substantially close to each other.

[0196] When the electrode plate **11C** passes between the pair of first upstream electrode plate

stopping rollers **348** and **349** substantially close to each other and the pair of first downstream electrode plate stopping rollers **353** and **354** substantially close to each other, and when the pair of first upstream electrode plate stopping rollers **348** and **349** and the pair of first downstream electrode plate stopping rollers **353** and **354** are stopped, a portion of the electrode plate **11C** between the first upstream electrode plate stopping rollers **348** and **349** and the first downstream electrode plate stopping rollers **353** and **354** stops as well. In this case, the first rotating head **310A** may attach the first tape piece **25A** to the first uncoated portion **16** of the electrode plate **11C**.

[0197] The second electrode plate stopping rollers **358**, **359**, **363**, and **364** stop the movement of the electrode plate **11C** passing through the second tape attachment point **TA2** when the second tape piece **25B** is attached to the electrode plate **11C**. The second electrode plate stopping rollers may include the second upstream electrode plate stopping rollers **358** and **359** arranged on an upstream side of the second electrode plate support **355** along the movement path of the electrode plate **11C** and the second downstream electrode plate stopping rollers **363** and **364** arranged on a downstream side of the second electrode plate support **355** along the movement path of the electrode plate **11C**.

[0198] The second upstream electrode plate stopping rollers **358** and **359** may be provided as, or include, a pair of second upstream electrode plate stopping rollers **358** and **359**, and the second downstream electrode plate stopping rollers **363** and **364** may be provided as, or include, a pair of second downstream electrode plate stopping rollers **363** and **364**. The pair of second upstream electrode plate stopping rollers **358** and **359** may be substantially close to each other, and the pair of second downstream electrode plate stopping rollers **363** and **364** may be substantially close to each other.

[0199] When the electrode plate **11C** passes between the pair of second upstream electrode plate stopping rollers **358** and **359** substantially close to each other and the pair of second downstream electrode plate stopping rollers **363** and **364** substantially close to each other, and when the pair of second upstream electrode plate stopping rollers **358** and **359** and the pair of second downstream electrode plate stopping rollers **363** and **364** are stopped, a portion of the electrode plate **11C** between the second upstream electrode plate stopping rollers **358** and **359** and the second downstream electrode plate stopping rollers **363** and **364** stops as well. In this case, the second rotating head **310B** may attach the second tape piece **25A** to the second uncoated portion **17** of the electrode plate **11C**.

[0200] The third electrode plate stopping rollers **368**, **369**, **373**, and **374** stop the movement of the electrode plate **11C** passing through the third tape attachment point **TA3** when the third tape piece **25C** is attached to the electrode plate **11C**. The third electrode plate stopping rollers may include the third upstream electrode plate stopping rollers **368** and **369** arranged on an upstream side of the third electrode plate support **365** along the movement path of the electrode plate **11C** and the third downstream electrode plate stopping rollers **373** and **374** arranged on a downstream side of the third electrode plate support **365** along the movement path of the electrode plate **11C**.

[0201] The third upstream electrode plate stopping rollers **368** and **369** may be provided as, or include, a pair of third upstream electrode plate stopping rollers **368** and **369**, and the third downstream electrode plate stopping rollers **373** and **374** may be provided as, or include, a pair of third downstream electrode plate stopping rollers **373** and **374**. The pair of third upstream electrode plate stopping rollers **368** and **369** may be substantially close to each other, and the pair of third downstream electrode plate stopping rollers **373** and **374** may be substantially close to each other.

[0202] When the electrode plate **11C** passes between the pair of third upstream electrode plate stopping rollers **368** and **369** that are substantially close to each other and the pair of third downstream electrode plate stopping rollers **373** and **374** that are substantially close to each other, and when the pair of third upstream electrode plate stopping rollers **368** and **369** and the pair of third downstream electrode plate stopping rollers **373** and **374** are stopped, a portion of the electrode plate **11C** between the third upstream electrode plate stopping rollers **368** and **369** and the

third downstream electrode plate stopping rollers **373** and **374** stops as well. In this case, the third rotating head **310C** may attach the third tape piece **25C** to the third uncoated portion **18** of the electrode plate **11C**.

[0203] The electrode plate tension rollers **376**, **382**, **386** and **392** pull the electrode plate **11C** so that tension of the electrode plate **11C** is maintained. The electrode plate tension rollers may include a first electrode plate tension roller **376** located on an upstream side of the first electrode plate support **345** and the first upstream electrode plate stopping rollers **348** and **349** along the movement path of the electrode plate **11C**, a second electrode plate tension roller **382** located between the first downstream electrode plate stopping rollers **353** and **354** and the second upstream electrode plate stopping rollers **358** and **359** along the movement path of the electrode plate **11C**, a third electrode plate tension roller **386** located between the second downstream electrode plate stopping rollers **363** and **364** and the third upstream electrode plate stopping rollers **368** and **369** along the movement path of the electrode plate **11C**, and a fourth electrode plate tension roller **392** located on a downstream side of the third electrode plate support **365** and the third downstream electrode plate stopping rollers **373** and **374** along the movement path of the electrode plate **11C**.

[0204] The electrode plate supply unit **340** may further include an upstream first electrode plate tension guide roller **378** on an upstream side of the first electrode plate tension roller **376** and a downstream first electrode plate tension guide roller **379** on a downstream side of the first electrode plate tension roller **376** along the movement path of the electrode plate **11C**.

[0205] The electrode plate **11C** supplied from the electrode plate supply roller **341** may sequentially pass through the upstream first electrode plate tension guide roller **378**, the first electrode plate tension roller **376**, the downstream first electrode plate tension guide roller **379**, and a gap between the pair of first upstream electrode plate stopping rollers **348** and **349**, and then move onto the first electrode plate support **345**.

[0206] In this section, when a tensile force of the electrode plate **11C** decreases, the first electrode plate tension roller **376** may move in a direction that increases the path of the electrode plate **11C**, to increase the tensile force of the electrode plate **11C**, and when the tensile force of the electrode plate **11C** increases, the first electrode plate tension roller **376** may move in a direction that decreases the path of the electrode plate **11C**, to decrease the tensile force of the electrode plate **11C**. Accordingly, the tensile force of the electrode plate **11C** supplied from the electrode plate supply roller **341** and moving to the first electrode plate support **345** may be maintained to be substantially uniform.

[0207] Since the first electrode plate tension roller **376** moves and the constant tensile force of the electrode plate **11C** can be maintained, the electrode plate supply roller **341** may substantially continuously rotate without stopping to attach the first tape piece **25A** to the electrode plate **11C** while the pair of first upstream electrode plate stopping rollers **348** and **349** and the pair of first downstream electrode plate stopping rollers **353** and **354** are stopped.

[0208] The electrode plate supply unit **340** may further include an upstream second electrode plate tension guide roller **383** on an upstream side of the second electrode plate tension roller **382** and a downstream second electrode plate tension guide roller **384** on a downstream side of the second electrode plate tension roller **382** along the movement path of the electrode plate **11C**.

[0209] In a section between the pair of first downstream electrode plate stopping rollers **353** and **354** and the pair of second upstream electrode plate stopping rollers **358** and **359**, when the tensile force of the electrode plate **11C** decreases, the second electrode plate tension roller **382** may move in a direction that increases the path of the electrode plate **11C**, to increase the tensile force of the electrode plate **11C**, and when the tensile force of the electrode plate **11C** increases, the second electrode plate tension roller **382** may move in a direction that decreases the path of the electrode plate **11C**, to decrease the tensile force of the electrode plate **11C**. Accordingly, the tensile force of the electrode plate **11C** moving from the electrode plate supply roller **345** toward the first electrode plate support **355** may be maintained to be substantially uniform.

[0210] The electrode plate supply unit **340** may further include an upstream third electrode plate tension guide roller **388** on an upstream side of the third electrode plate tension roller **386** and a downstream third electrode plate tension guide roller **389** on a downstream side of the third electrode plate tension roller **386** along the movement path of the electrode plate **11C**.

[0211] In a section between the pair of second downstream electrode plate stopping rollers **363** and **364** and the pair of third upstream electrode plate stopping rollers **368** and **369**, when the tensile force of the electrode plate **11C** decreases, the third electrode plate tension roller **386** may move in a direction that increases the path of the electrode plate **11C**, to increase the tensile force of the electrode plate **11C**, and when the tensile force of the electrode plate **11C** increases, the third electrode plate tension roller **386** may move in a direction that decreases the path of the electrode plate, to decrease the tensile force of the electrode plate **11C**. Accordingly, the tension force of the electrode plate **11C** moving from the second electrode plate support **355** toward the third electrode plate support **365** may be maintained to be substantially uniform.

[0212] The electrode plate supply unit **340** may further include an upstream fourth electrode plate tension guide roller **393** on an upstream side of the fourth electrode plate tension roller **392**, and a downstream fourth electrode plate tension guide roller **394** on a downstream side of the fourth electrode plate tension roller **392** along the movement path of the electrode plate **11C**.

[0213] The electrode plate **11C**, on which the third tape piece **25C** is attached to the third uncoated portion **18**, may pass between the pair of third downstream electrode plate stopping rollers **373** and **374**, may sequentially pass through the upstream fourth electrode plate tension guide roller **393**, the fourth electrode plate tension roller **392**, and the downstream fourth electrode plate tension guide roller **394**, and may be collected by the electrode plate collection roller **343**.

[0214] In this section, when the tensile force of the electrode plate **11C** decreases, the fourth electrode plate tension roller **392** may move in a direction that increases the path of the electrode plate **11C**, to increase the tensile force of the electrode plate **11C**, and when the tensile force of the electrode plate **11C** increases, the fourth electrode plate tension roller **392** may move in a direction that decreases the path of the electrode plate **11C**, to decrease the tensile force of the electrode plate **11C**. Accordingly, the tension force of the electrode plate **11C** moving from the third electrode plate support **365** toward the electrode plate collection roller **343** may be maintained to be substantially uniform.

[0215] Because the fourth electrode plate tension roller **392** moves and the constant tensile force of the electrode plate **11C** can be maintained, the electrode plate collection roller **343** may substantially continuously rotate without stopping to attach the third tape piece **25C** to the electrode plate **11C** while the pair of third upstream electrode plate stopping rollers **368** and **369** and the pair of third downstream electrode plate stopping rollers **373** and **374** are stopped.

[0216] The first rotating head **310A**, which adsorbs the first tape piece **25A**, may move to the first tape attachment point **TA1**, contemporaneously or at the same time, the second rotating head **310B**, which adsorbs the second tape piece **25B**, may move to the second tape attachment point **TA2**, and the third rotating head **310C**, which adsorbs the third tape piece **25C**, may move to the third tape attachment point **TA3**.

[0217] In a state in which the first rotating head **310A** and the first uncoated portion **16** of the electrode plate **11C** are aligned in the radial direction of the rotation center **RX**, the first rotating head **310A** may move in the radial direction from the rotation center **RX** to attach the first tape piece **25A** to the first uncoated portion **16** of the electrode plate **11C**. The first tape piece **25A** may be attached to the electrode plate **11C** to cover the first uncoated portion **16**.

[0218] In a state in which the second rotating head **310B** and the second uncoated portion **17** of the electrode plate **11C** are aligned in the radial direction of the rotation center **RX**, the second rotating head **310B** may move in the radial direction from the rotation center **RX** to attach the second tape piece **25B** to the second uncoated portion **17** of the electrode plate **11C**. The second tape piece **25B** may be attached to the electrode plate **11C** to cover the second uncoated portion **17**.

[0219] In a state in which the third rotating head **310C** and the third uncoated portion **18** of the electrode plate **11C** are aligned in the radial direction of the rotation center **RX**, the third rotating head **310C** may move in the radial direction from the rotation center **RX** to attach the third tape piece **25C** to the third uncoated portion **18** of the electrode plate **11C**. The third tape piece **25C** may be attached to the electrode plate **11C** to cover the third uncoated portion **18**.

[0220] In various examples, the rotating heads **310A**, **310B**, and **310C** may stop adsorbing the tape pieces **25A**, **25B**, and **25C** and return toward the rotation center **RX**. After the first tape piece **25A** is attached to the electrode plate **11C**, the first rotating head **310A** may further rotate in the same rotation direction as a case in which the first rotating head **210A** rotates to the first tape attachment point **TA1** to move toward the first tape transfer point **TF1**.

[0221] After the second tape piece **25B** is attached to the electrode plate **11C**, the second rotating head **310B** may further rotate in the same rotation direction as a case in which the second rotating head **310B** rotates to the second tape attachment point **TA2** to move toward the second tape transfer point **TF2**. After the third tape piece **25C** is attached to the electrode plate **11C**, the third rotating head **310C** may further rotate in the same rotation direction as a case in which the third rotating head **310C** rotates to the third tape attachment point **TA3** to move toward the third tape transfer point **TF3**.

[0222] Although not illustrated in FIGS. **11** and **12**, the electrode tab welding unit, which welds the electrode tab (not illustrated) to the uncoated portions **16**, **17** and **18** of the electrode plate **11C**, may be located on the movement path of the electrode plate **11C** from the electrode plate supply roller **341** toward the first electrode plate tension roller **376**. In this case, at the tape attachment points **TA1**, **TA2** and **TA3**, the tape pieces **25A**, **25B**, and **25C** may be attached to the uncoated portions **16**, **17**, and **18** to cover the uncoated portion **16**, **17**, and **18** along with the portion of the electrode tab welded to the uncoated portion **16**, **17** and, **18**.

[0223] In the tape attaching apparatus **300** according to the example embodiment of the present disclosure, the numbers of the plurality of tape transfer points **TF1**, **TF2**, and **TF3** and the tape attachment points **TA1**, **TA2**, and **TA3** are the same, and the number of rotating heads **310A**, **310B**, and **310C** is twice the number of tape transfer points **TF1**, **TF2**, and **TF3** and the number of tape attachment points **TA1**, **TA2**, and **TA3**.

[0224] The tape pieces **25A**, **25B**, **24C** are transferred to one of the plurality of the rotating heads **310A**, **310B**, and **310C** from the tape supply units **320A**, **320B**, and **320C** at one of the plurality of tape transfer points **TF1**, **TF2**, and **TF3** and are transferred and attached to the electrode plate **11C** from the one rotating head at one of the plurality of tape attachment points **TA1**, **TA2**, and **TA3**. The one tape transfer point and the one tape attachment point are spaced apart from each other at about 180° intervals with respect to the plurality of rotation centers **RX**.

[0225] A size of the tape pieces **25A**, **25B**, and **25C** transferred to one of the plurality of rotating heads **310A**, **310B**, and **310C** at one of the plurality of tape transfer points **TF1**, **TF1** and **TF2** is different from a size of the tape pieces **25A**, **25B**, and **25C** transferred to another of the plurality of rotating heads **310A**, **310B**, and **310C** at another of the plurality of tape transfer points **TF1**, **TF2** and, **TF3**.

[0226] The plurality of tape supply units **320A**, **320B**, and **320C** are provided in a one-to-one correspondence with the tape transfer points **TF1**, **TF2**, and **TF3**, and the one electrode plate supply unit **340** is provided.

[0227] The plurality of tape attachment points **TA1**, **TA2**, and **TA3** are provided, the one electrode plate supply roller **341** is provided, and the plurality of electrode plate supports **345**, **355**, and **365** are provided in a one-to-one correspondence with the tape attachment points **TA1**, **TA2**, and **TA3**. The plurality of electrode plate supports **345**, **355** and **365** are spaced apart from each other on the movement path of the electrode plate **11C** supplied from the electrode plate supply roller **341**.

[0228] According to an example embodiment of the present disclosure, a head that receives a tape and attaches the tape to an electrode plate does not move in a straight line but rotates. Accordingly,

a tape supply unit and an electrode plate supply unit can be arranged around the rotating head, and thus a small size of a tape attaching apparatus is easily designed. Therefore, an installation space of a tape attaching apparatus can be reduced or minimized, manufacturing costs of a secondary battery can be reduced, and accuracy of a tape attachment position can be improved.

[0229] According to example embodiments of the present disclosure, a plurality of rotating heads are arranged to rotate about the same rotation center in one direction, and thus productivity of tape attaching work can be easily improved.

[0230] However, aspects and features of the present invention are not limited to those described above and other aspects and features not mentioned will be clearly understood by those skilled in the art from the detailed description given above.

[0231] Although the present invention has been described with reference to some example embodiments and drawings illustrating aspects thereof, the present invention is not limited thereto. Various modifications and variations can be made by a person skilled in the art to which the present invention belongs within the scope of the technical spirit of the invention and the claims and equivalents thereto.

Claims

1. A tape attaching apparatus comprising: a tape supply unit configured to supply a tape; an electrode plate supply unit configured to supply an electrode plate; and a rotating head configured to receive the tape supplied from the tape supply unit and to rotate so as to attach the tape to the electrode plate supplied from the electrode plate supply unit.
2. The tape attaching apparatus as claimed in claim 1, wherein the rotating head is configured to receive the tape from the tape supply unit at a tape transfer point, and is configured to rotate in one direction to attach the tape to the electrode plate at a tape attachment point.
3. The tape attaching apparatus as claimed in claim 2, wherein, the rotating head is further configured to rotate in the one direction to return to the tape transfer point when the tape is attached to the electrode plate at the tape attachment point.
4. The tape attaching apparatus as claimed in claim 2, wherein the tape includes an adhesive surface coated with an adhesive, and a non-adhesive surface which is opposite to the adhesive surface and is not coated with the adhesive, and the rotating head is configured to adsorb the non-adhesive surface at the tape transfer point and to stop the adsorption of the non-adhesive surface at the tape attachment point.
5. The tape attaching apparatus as claimed in claim 4, wherein the rotating head is configured to push the tape toward the electrode plate to facilitate adhesion of the adhesive surface to the electrode plate at the tape attachment point.
6. The tape attaching apparatus as claimed in claim 2, wherein the rotating head comprises a plurality of rotating heads, and the plurality of rotating heads are arranged at equiangular intervals with respect to the same rotation center.
7. The tape attaching apparatus as claimed in claim 6, further comprising a rotation frame configured to support the plurality of rotating heads, the rotation frame being configured to rotate about the rotation center.
8. The tape attaching apparatus as claimed in claim 2, wherein the tape transfer point comprises a plurality of tape transfer points, the tape attachment point comprises a plurality of tape attachment points, and the number of tape transfer points is equal to the number of tape attachment points, and a number of rotating heads is twice the number of tape transfer points and the number of tape attachment points.
9. The tape attaching apparatus as claimed in claim 8, wherein the tape is transferred to one of the plurality of rotating heads from the tape supply unit at one of the plurality of tape transfer points, transferred to the electrode plate from the one rotating head at one of the plurality of tape

attachment points, and attached to the electrode plate, and the tape transfer point and the tape attachment point are spaced apart from each other by about 180° with respect to a rotation center of the rotating head.

10. The tape attaching apparatus as claimed in claim 8, wherein a size of the tape transferred to one of the plurality of rotating heads at one of the plurality of tape transfer points is different from a size of the tape transferred to another of the plurality of rotating heads at another of the plurality of tape transfer points.

11. The tape attaching apparatus as claimed in claim 8, wherein the tape supply unit comprises a plurality of tape supply units in a one-to-one correspondence with the tape transfer points, and the electrode plate supply unit comprises one electrode plate supply unit.

12. The tape attaching apparatus as claimed in claim 1, wherein the tape supply unit comprises: a tape supply roller configured to substantially continuously supply the tape; and a tape cutter configured to cut the tape supplied from the tape supply roller to a preset length.

13. The tape attaching apparatus as claimed in claim 12, wherein the tape supply unit further comprises a tape gripper configured to grip an end portion of the tape fed from the tape supply roller.

14. The tape attaching apparatus as claimed in claim 12, wherein, the rotating head is configured to adsorb the tape when the tape gripper grips the tape, and the tape cutter is configured to cut the tape when the tape is adsorbed to the rotating head.

15. The tape attaching apparatus as claimed in claim 14, wherein the tape supply unit further comprises a tape tension roller configured to pull the tape so that tension of the tape is substantially maintained.

16. The tape attaching apparatus as claimed in claim 2, wherein the electrode plate supply unit comprises: an electrode plate supply roller configured to substantially continuously supply the electrode plate; and an electrode plate support configured to support the electrode plate at the tape attachment point.

17. The tape attaching apparatus as claimed in claim 16, wherein the electrode plate supply unit further comprises an electrode plate stopping roller configured to stop movement of the electrode plate passing through the tape attachment point when the tape is attached to the electrode plate.

18. The tape attaching apparatus as claimed in claim 17, wherein the rotating head is configured to rotate after adsorbing the tape at the tape transfer point, to move in a radial direction from a rotation center of the rotating head at the tape attachment point to attach the tape to the electrode plate, to stop the adsorbing of the tape, and to return toward the rotation center.

19. The tape attaching apparatus as claimed in claim 17, wherein the electrode plate supply unit further comprises an electrode plate tension roller configured to pull the electrode plate so that tension of the electrode plate is substantially maintained.

20. The tape attaching apparatus as claimed in claim 16, wherein the tape attachment point comprises a plurality of tape attachment points, the electrode plate supply roller comprises one electrode plate supply roller, the electrode plate support comprises a plurality of electrode plate supports in a one-to-one correspondence with the tape attachment points, and the plurality of electrode plate supports are spaced apart from each other on a movement path of the electrode plate supplied from the electrode plate supply roller.
